Superpixel-Based Collaborative and Low-Rank Regularization for Sparse Hyperspectral Unmixing

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***Abstract*— Sparse unmixing (SU) has been widely applied to remotely sensed hyperspectral images (HSIs) interpretation. Compared with traditional unmixing algorithms, SU does not need to extract pure signatures (endmembers) from the image. The endmember matrix is constructed by directly selecting spectra from a known library, which is used to estimate the fractional abundances associated with endmembers. This avoids the problem of extracting virtual endmembers without physical meaning. However, SU does not generally include spatial information, which may limit its performance. In order to address this limitation and include local spatial information, low-rank and sparse features in local regions can be exploited. In this article, we include spatial information in the traditional SU algorithm by extracting low-rank and spatial information based on superpixels and further propose an algorithm named superpixel-based collaborative sparse and low-rank regularization for SU (SCLRSU) to improve the performance of the traditional spatial regularization-based SU methods. In our proposed method, we combine superpixel segmentation and structural sparsity. Experiments are carried out on two simulated datasets and two real HSI datasets, and our results are compared with those obtained by traditional SU methods. Our results indicate that our newly proposed method provides very competitive performance.**

***Index Terms*— Hyperspectral remote sensing, low rank, sparse unmixing (SU), superpixel segmentation.**

I. Introduction

With the spectral imaging technology progressing rapidly, hyperspectral remotely sensed data have been widely used in urban construction [1], military reconnaissance [2], mineral identification [3], change detection [4], [5], geographic mapping [6], and object recognition [7]. However, because of the spatial resolution limitation of hyperspectral image (HSI) sensors and the complexity and diversity of ground objects, mixed pixels have a great impact on HSI data analysis and application. For the purpose of addressing the problem of mixed pixels, hyperspectral unmixing techniques have been proposed to decompose HSI data into a few different types of surface materials, named endmembers [8], and their corresponding proportions, named abundances [9].

Currently, most hyperspectral unmixing methods are based on the linear mixture model (LMM) [10]–[12] and the nonlinear mixture model (NLMM) [13]–[15]. Although the research on NLMM is becoming popular, the application of NLMM is limited due to its many parameters, complexity, and requirement of prior information, while the application of LMM is wider due to advantages such as simplicity, high efficiency, and clear analytical significance.

The early methods based on LMM are mainly based on statistical and geometric unmixing [16]–[18]. These methods are pure pixel-based methods that can directly extract endmembers from HSIs, including the pixel purity index (PPI) [19], the vertex component analysis (VCA) [20], and the simplex growing algorithm (SGA) [21]. However, the pure pixel assumption is difficult to realize in real data, and estimating the number of endmembers can be a great challenge [22].

In order to solve and overcome the above drawbacks, sparse unmixing (SU) [23] has been developed based on the compressed sensing and sparse representation theory [24], [25]. The SU method does not need endmember extraction. It assumes that the mixed pixels are composed of a variety of pure endmembers and relies on a known spectral library as prior knowledge to avoid the influence of incorrect endmember extraction on unmixing accuracy effectively. For instance, SU by variable splitting and augmented Lagrangian (SUnSAL) algorithm [26] is a typical SU method. It used the L_1 norm, addressing the shortcomings of classical unmixing algorithms, i.e., it does not need to estimate the number of endmembers or to extract their signatures from the data. Since the row sparsity of the matrix can be accurately measured by the L_{21} norm, the collaborative SUnSAL (CLSUnSAL) [27] was developed to improve the global sparsity and make full use of the similarity of endmembers in local regions of the image by introducing a norm constraint to the abundance matrix. In other words, CLSUnSAL uses the L_{21} norm to assure that

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all the abundance vectors have global row sparsity. As a result, the model is robust to outliers by sharing the same sparse structure. However, CLSUnSAL ignores the spatial information contained in the HSI, which reduces the unmixing accuracy [28]. When the endmembers are only distributed in the local homogeneous regions rather than in the whole HSI, the efficiency of this method decreases. To include the local spatial information, total variation (TV) spatial regularization uses the correlation between endmembers of adjacent pixels and abundance information to include the local spatial information, and it can significantly improve the decomposition results. The TV-regularized variable separation and augmented Lagrangian algorithm (SUnSAL-TV) [29] introduced a TV spatial regularization term into the SUnSAL to smooth the abundance matrix and make better use of the spatial information contained in HSIs [11], [30]–[32]. Furthermore, Zhang *et al.* [33] proposed a decomposition model, named spectral–spatial weighted SU (S^2WSU), with only one regularization term. An iterative weighting method is used in the spectral weighting factor for the enhancement of the sparsity of endmembers in the spectral library, while the reciprocal of the local block sum of the adjacency matrix is used in the spatial weighting factor to obtain similar abundance values in the blocks [34]. However, the unmixing accuracy of S^2WSU needs to be improved under high noise conditions. Recently, new sparse decomposition methods based on uniform regions have been developed. For example, Zhang *et al.* [32] proposed a local cooperative SU (LCSU), which performs LCSU in a local region with fixed window size. LCSU uses similarity of spectral and proximity of location (local similarity) to capture spatial prior information. Qi *et al.* [35] developed a new spectral–spatial-weighted multiview collaborative SU (MCSU) model, which uses a weighted sparse regularizer including both multiview spectral and spatial weighting factors to further impose sparsity on the fractional abundances. Das *et al.* [36] proposed a novel, semisupervised, parameter-free algorithm that employs sparsity measures for library pruning; it includes sparsity criteria-based library pruning and a sparse inversion method for abundance computation. Miao and Yang [37] proposed an unsupervised SU method using comprehensive-learning-based particle swarm optimization (PSO), which leads to dealing with the implementation of unmixing in refined solution spaces and high-dimensional problems. Das *et al.* [38] proposed a mutual coherence reduction method that can identify the actual image endmembers utilizing the low-rank constraint. It employed a dictionary pruning step to improve the performance of library-aided unmixing. Li *et al.* [39] established a new technique called superpixel-based low-rank SU (SpLRSU), which uses superpixel segmentation to adaptively generate local homogeneous regions and enforced the low-rank constraint to preserve the low-dimensional structure of superpixel blocks.

In recent years, it has been observed that low-rank constraints can extract the low-dimensional structure information of HSIs [40]. Due to the fact that the spectral characteristics of local pixels in HSIs are similar, low-rank constraints have been widely applied in the fields of HSI processing, for example, denoising [41], detection [42], classification [43], and unmixing [44]–[46]. Giampouras *et al.* [47] simultaneously imposed both low rank and sparse on the entire abundance matrix. Zhang *et al.* [48] designed a spectral library pruning and low-rank representation strategy for unmixing. Their method assumes that the abundance coefficients of adjacent pixels with the same endmember are similar, and if the abundance coefficient of an endmember does not contain any value greater than the threshold, it is not a real endmember and should be deleted from the spectral library. Hong *et al.* [49] proposed a low-rank attribute embedding method based on subspaces. Li *et al.* [50] proposed a novel SU algorithm named superpixel-based reweighted low-rank and TV. It uses superpixel segmentation technology to design adaptive neighborhoods with more accurate spatial–contextual information. In addition, it enforced a low-rank constraint on the objective function and reweighted nuclear norm to achieve a more efficient and accurate low-rank constraint. The low-rank property of HSIs is embedded into the original subspace, and the spectral variability in the process of unmixing is solved by projecting the original data into the low-rank subspace.

However, the aforementioned unmixing methods do not make full use of spectral and spatial information, and their performance is in need of improvement. In spatial regions, pixels in local areas usually contain the same (or similar) characteristics. In order to solve the above problem, we propose a new superpixel-based collaborative sparse and low-rank regularization for SU (SCLRSU). First, the HSI is segmented using superpixels. Then, structural sparsity is taken as the regularization constraint to improve the traditional hyperspectral unmixing algorithm, and both spectral and spatial weightings are applied to each superpixel.

In order to evaluate the effectiveness of our newly proposed method SCLRSU, five typical sparse methods, SUnSAL [26], CLSUnSAL [27], SUnSAL-TV [29], S^2WSU [33], and SUSRLR-TV [50], are compared to our newly proposed method. We use two different synthetic datasets and two real datasets. Our experimental results reveal that the proposed SCLRSU method can integrate spatial information and spectral information effectively and improve the performance of hyperspectral unmixing. The proposed unmixing algorithm exhibits the following innovative features.

1) It combines local spatial and spectral information. First, superpixel segmentation is implemented to obtain homogeneous regions. Then, a collaborative low-rank constraint is applied to keep the low-dimensional structure of the abundance matrix associated with each superpixel. The two constraints are incorporated into the SU framework and effectively addressed by the alternating direction method of multipliers (ADMM) method [26].

2) It synchronously exploits the local spatial smoothness and global row sparsity via collaborative sparsity and low-rank regularizations, suppressing noise and keeping the abundance structural information.

The rest of this article is organized as follows. Section II outlines the basic principles of SU. Section III detailedly describes the proposed SCLRSU method. Section IV describes the experimental dataset and analyzes the obtained results.

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Section V concludes this article with some remarks and hints at plausible future research lines.

II. Related Work

A. Linear Mixture Model

The LMM is a widely used hyperspectral unmixing method, which assumes the HSI can be expressed as a linear combination of a few pure spectral features [51]. In the LMM, each pixel can be divided into several objects

$$Y = MX + N \tag{1}$$

where Y is an HSI composed of l bands and n pixels, M is an $l \times n$ spectral dictionary matrix, X is an $m \times n$ abundance matrix, and N is an $l \times n$ independent random Gaussian noise matrix.

In order to assure that the abundance solution has a physical meaning, two constraints are included, called abundance nonnegativity constraint (ANC) [52] and abundance sum-to-one constraint (ASC) [53]

$$\text{ANC} = \alpha_i \geq 0 \tag{2}$$

$$\text{ASC} = \sum_{i=1}^{q} \alpha_j = 1. \tag{3}$$

While the ANC is commonly adopted in SU, the ASC is generally prone to some criticisms related to the difficulty to obtain a full set of endmembers in many applications [26].

B. Sparse Unmixing

Sparse decomposition extracts spectral signatures from a large spectral library $A \in R^{l \times t}$, where t is the number of ground materials in spectral library A. The method can be expressed as follows:

$$Y = AX + N \tag{4}$$

where $x \in R^{l \times t}$ is the matrix of abundance in the spectral library A. Normally, in a given real hyperspectral scene, the number of endmembers is usually much less than the spectral features in A. Assuming that the HSI contains P endmembers, the abundance matrix X of the solution should be sparse because the number of atoms p in A is much larger than the number of endmembers $p \gg P$ in the HSI. Based on this theoretical analysis, we can narrow the selection range to a satisfactory solution by adding sparse constraints. The process of adding sparse constraints is called regularization. In summary, there are many zero values in the abundance matrix X, so it is considered to be sparse. The sparse decomposition model can be written as

$$\min \frac{1}{2}||AX - Y||_F^2 + \lambda||X||_0 \quad \text{s.t } X \geq 0 \tag{5}$$

where $||AX - Y||_F^2$ is the model reconstruction error. $||X||_F = \sqrt{\text{trace}\{XXT\}}$ is the Frobenius norm of X, $||X||_0$ is the L_0 norm of X. The minimization of (5) is difficult to solve. It has been shown, however, that when A satisfies the restricted isometric property (RIP) [54], [55], the L_0 problem can be transformed into an L_1 problem, which is a convex optimization one and is easier to solve. Thus, we can use the L_1 norm instead of the L_0 norm. Equation (6) can be rewritten as follows:

$$\min \frac{1}{2}||AX - Y||_F^2 + \lambda||X||_1 \quad \text{s.t } X \geq 0 \tag{6}$$

where $||X||_1 = \sum_{i=1}^{t}\sum_{j=1}^{s}|x_{ij}|$ is L for calculating the sum of elements in the L_1 norm of A.

HSIs usually contain only several endmembers from a large spectral library. The CLSUnSAL algorithm forces the column pixels of X to share the same active endmember sets, so X has only several nonzero rows. In other words, X is sparse between lines. To take advantage of this prior knowledge, CLSUnSAL uses the L_{21} norm instead of the L_1 norm as follows:

$$\min \frac{1}{2}||AX - Y||_F^2 + \lambda||X||_{21} + l_{R_+}(X) \quad \text{s.t } X \geq 0 \tag{7}$$

where the abundance matrix $||X||_{21} = \sum_{i=1}^{t}\sum_{j=1}^{s}|x_{ij}|$ represents the ith line of X.

In addition to the use of spectral characteristics, the correlation of spatial information of HSIs can also improve the unmixing effect. Each pixel and its adjacent pixels usually contain similar endmembers, so the spectral characteristics between these pixels are very similar. In order to make improvement of the smoothness between the abundance estimations associated with spatially adjacent pixels, TV regularization includes spatial information as follows:

$$\min \frac{1}{2}||AX - Y||_F^2 + \lambda||X||_1 + \lambda_{\text{TV}}\text{TV}(X) + l_{R_+}(X) \tag{8}$$

where λ_{TV} is the parameter of TV. The SUnSAL-TV method uses anisotropy. The TV can be simply defined as

$$||X||_{\text{TV}} = ||HX||_1 \tag{9}$$

where H is made of two difference operators named H_h and H_v. H_h calculates the horizontal differences of adjacent pixels, while H_v calculates the vertical differences of adjacent pixels

$$H = [H_h; H_v]. \tag{10}$$

III. Proposed Method

In order to obtain the local information and make full use of the local similarity of HSIs, our newly developed SCLRSU uses superpixel segmentation to accurately obtain homogeneous regions. In the following, we describe the different steps adopted by our method.

A. Superpixel Segmentation

The simple linear iterative clustering (SLIC) algorithm is used for superpixel segmentation in this work [56]. It was originally used for color image superpixel segmentation. The main idea is to comprehensively consider similar texture, color, brightness, spatial information, and other features and use their similarities to group the pixels. Finally, the image is divided into several separate regions with high similarity. Superpixel segmentation is simple, with few parameters, and fast. This algorithm only needs two parameters: Q and Ws, where Q represents the number of superpixels to be segmented, and Ws

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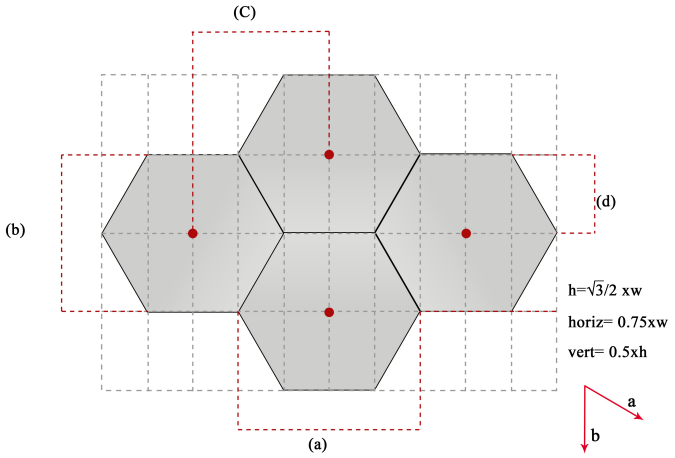


Fig. 1. Metrics of flat-topped hexagonal grids. (a) w. (b) h. (c) Horiz. (d) Vert.

represents an index between the spatial and spectral distances of each pixel in the cluster.

In order to improve the consistence of superpixel segmentation, the initial clustering is generated by using hexagonal grids instead of square grids in the standard SLIC algorithm. As shown in Fig. 1, each side and corner of a hexagon are shared by two and three hexagons, respectively [57]. In addition, its size is described by w (width) and h (height), and the spacing between adjacent hexagons can be expressed by the horizontal distance W and the vertical distance Hh.

Compared with the square mesh, the hexagonal mesh has two obvious advantages in terms of image segmentation: 1) each hexagon has more diagonal neighborhood than the square and 2) the hexagonal mesh provides fewer distance distortions for boundary pixels [58]. The calculation of the face center of a hexagon is a little more complicated than that of a square, which can be described as a simple matrix multiplication

$$\begin{bmatrix} m^i \\ n^i \end{bmatrix} = \begin{bmatrix} \text{vert} & 0 \\ 0 & \text{horiz} \end{bmatrix} \begin{bmatrix} 2\text{row}_i - 1 \\ \text{col}_i \end{bmatrix} \tag{11}$$

where row_i and col_i is the index of the row and column of the ith superpixel.

Besides the initial clustering, similar pixels will be searched in the area of $2w * 2w$ around the center of the superpixel. First, the dimension of the HSI will be reduced, and then, the reduced data will be segmented. At this time, the HSI will be divided into homogeneous regions with different sizes and shapes, and each homogeneous region is called a superpixel. For an HSI with S pixels, let us suppose that s blocks of superpixels are obtained by using the superpixel segmentation method, and the endmember distribution of each area in these superpixels is consistent. Because the endmember of the whole image is limited, there will be a large number of regions with the same endmember distribution in these s homogeneous regions, and each superpixel will contain s/S pixels. Different from 3-D color images, in HSIs, the spectral distance measurement is measured by using spectral angle distance (SAD), and the spatial (Euclidean) distance from each pixel to the cluster center needs to be calculated in the process of clustering. In our context, the color distance dc, the spatial distance ds, and the metric distance D are expressed as follows:

$$dc = \sqrt{(l_1 - c_1)^2 + (l_2 - c_2)^2 + \cdots + (l_{10} - c_{10})^2} \tag{12}$$

$$ds = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \tag{13}$$

$$D = \sqrt{\left(\frac{d_c}{N_c}\right)^2 + \left(\frac{d_s}{N_s}\right)^2} \tag{14}$$

where dc is the color distance, ds is the spatial distance, and Ns is the maximum space distance within the class.

The value of k is very important for the algorithm. If the value of k is too large, regions containing a small number of pixels may be generated, which is not appropriate to exploit the correlation between abundance vectors. If we set k too small, some unrelated pixels may be included in the same superpixel. Using these abundance vectors with poor correlation may result in poor unmixing performance. The concrete steps of SLIC are given in Algorithm 1.

Algorithm 1 SLIC for HSI Segmentation

Input: hyperspectral data, Y: number of superpixels k, iteration times i;
Output: Segmentation result;
Repeat
1. According to the hexagon width w and height h, the image is segmented initially. The initial coordinate is the center pixel, and the average spectrum is the center pixel spectrum.
2. Calculate the distance d from all pixels within distance $2w * 2w$ of each cluster center to the cluster center.
3. Update the center coordinates and the center spectrum of the superpixel. The center coordinates are the average coordinates of all the pixels in the superpixel, and the center spectrum is the average spectrum of all the pixels in the superpixel.
Stop

B. Cooperative Low-Rank Constraint

Since the abundance image has the same high spatial and spectral correlation as the HSI, it is necessary to transform the segmented abundance matrix into 3-D image form, that is, $X \in^{nc,ns,nr}$. Then, in terms of the superpixel segmentation result, the 3-D abundance data of the corresponding position are extracted, and S superpixels related to the abundance data are obtained, in which the pixels' number in each superpixel is related to the size of the superpixel. Let r be $Xr \in R^{\text{sn}*\text{tr}}$, where $r = 1, 2, \ldots, R$, sn is the number of pixels in the rth superpixel, and t is the number of spectral bands. Due to the fact that the spectra of the homogeneous region obtained by each superpixel have similarities, and only a small part of the spectral signatures participate in the decomposition, the superpixels satisfy the sparse constraint. Therefore, the method in this article uses collaborative sparsity to force all pixels in

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the superpixel to share the same endmember subset. It can be written as follows:

$$\|X_r\| = \sum_{t=1}^{\mathrm{sn}} \sqrt{\sum_{j=1}^{t} |X_r^{ij}|^2} \tag{15}$$

where X_r^{ij} denotes the superpixel x. In addition, the spatial distribution of pixels in a superpixel has a high linear correlation, which means that the pixels in the homogeneous region have low-rank properties. After adding the low-rank constraint, it can be expressed as

$$\|X_r\|_{1/2,*} = \sum_{i=1}^{\min(\mathrm{sn},t)} \left(\beta_i^{1/2}(\|X_r\|)\right)^2 \tag{16}$$

where $\beta_i^{1/2}(X_r)$ indicates that X_r is a singular value of R.

The method in this article applies the collaborative sparse and low-rank regularization terms to each superpixel and uses the spatial–spectral double-weighted constraint term to extract the spatial information and spectral information of the pixel for the abundance matrix [33]. Based on the aforementioned analysis, we can build the final objective function as follows:

$$\min \frac{1}{2}||AX - Y||_F^2 + \lambda \left\| \left(W_{\mathrm{spe}} W_{\mathrm{spa}}\right) V_2 \right\|_{2,1} + \lambda_{\mathrm{cs}} \sum_{r=1}^{R} \|X_r\|_{1/2,*} + l_{R_+}(X) \tag{17}$$

where λ and λ_{cs} are the parameters of spatial–spectral weighting and collaborative sparse low-rank regularization terms. W_{spe} are the sparsity of the endmembers in the spectral library. W_{spa} is the spatial weighting factor [33]. $l_{R+}(X)$ is an indicator function used to express ANC constraint. The ADMM is chosen to solve problem (17) in the end. Algorithm 2 shows the concrete steps of the SCLRSU. More details can be referred to the Appendix.

Algorithm 2 SCLRSU for HSI Unmixing

Initialization: k, t=0, input μ, λ, λcs, $\mathrm{V}_1, \mathrm{V}_2, \mathrm{V}_{3,\mathrm{r}}, \mathrm{V}_4, \mathrm{D}_1, \mathrm{D}_2, \mathrm{D}_{3,\mathrm{r}}, \mathrm{D}_4$

Iterate the algorithm until the conditions are met, otherwise end the algorithm:

1. **Update** $\mathrm{W}_{(\mathrm{t})}^{\mathrm{spe}}$
2. **Update** $\mathrm{W}_{(\mathrm{t})}^{\mathrm{spa}}$
3. Iterate the algorithm until the conditions are met, otherwise end the algorithm:

$\mathrm{argmin}_{\mathrm{X},\mathrm{V}_1,\mathrm{V}_2,\mathrm{V}_{3,\mathrm{r}},\mathrm{V}_4,} = \mathrm{argminL}(\mathrm{V}_1, \mathrm{V}_2, \mathrm{V}_{3,\mathrm{r}}, \mathrm{V}_4, \mathrm{D}_1, \mathrm{D}_2, \mathrm{D}_{3,\mathrm{r}}, \mathrm{D}_4)$

4. Update

$\mathrm{V}_1^{(\mathrm{k}+1)} = \mathrm{argminL}(\mathrm{V}_1, \mathrm{V}_2, \mathrm{V}_{3,\mathrm{r}}, \mathrm{V}_4, \mathrm{D}_1, \mathrm{D}_2, \mathrm{D}_{3,\mathrm{r}}, \mathrm{D}_4)$

$\mathrm{V}_2^{(\mathrm{k}+1)} = \mathrm{argminL}(\mathrm{V}_1, \mathrm{V}_2, \mathrm{V}_{3,\mathrm{r}}, \mathrm{V}_4, \mathrm{D}_1, \mathrm{D}_2, \mathrm{D}_{3,\mathrm{r}}, \mathrm{D}_4)$

$\mathbf{V}_{\mathbf{3,r}}^{(\mathrm{k}+1)} = \mathrm{argminL}(\mathrm{V}_1, \mathrm{V}_2, \mathrm{V}_{3,\mathrm{r}}, \mathrm{V}_4, \mathrm{D}_1, \mathrm{D}_2, \mathrm{D}_{3,\mathrm{r}}, \mathrm{D}_4)$

$\mathrm{V}_{\mathbf{4}}^{(\mathrm{k}+1)} = \mathrm{argminL}(\mathrm{V}_1, \mathrm{V}_2, \mathrm{V}_{3,\mathrm{r}}, \mathrm{V}_4, \mathrm{D}_1, \mathrm{D}_2, \mathrm{D}_{3,\mathrm{r}}, \mathrm{D}_4)$

5. Update multiplier:

$\mathrm{D}_1^{(\mathrm{k}+1)} = \mathrm{D}_1^{(\mathrm{k}+1)} - \mathrm{AX}^{(\mathrm{k}+1)} + \mathrm{V}_1^{(\mathrm{k}+1)}$

$\mathrm{D}_2^{(\mathrm{k}+1)} = \mathrm{D}_2^{(\mathrm{k}+1)} - \mathrm{X}^{(\mathrm{k}+1)} + \mathrm{V}_2^{(\mathrm{k}+1)}$

$\mathrm{D}_{3,\mathrm{r}}^{(\mathrm{k}+1)} = \mathrm{D}_{3,\mathrm{r}}^{(\mathrm{k}+1)} - \mathrm{X}^{(\mathrm{k}+1)} + \mathrm{V}_{3,\mathrm{r}}^{(\mathrm{k}+1)}$

$\mathrm{D}_4^{(\mathrm{k}+1)} = \mathrm{D}_4^{(\mathrm{k}+1)} - \mathrm{X}^{(\mathrm{k}+1)} + \mathrm{V}_4^{(\mathrm{k}+1)}$

6. Update iteration:

$\mathrm{k} \leftarrow \mathrm{k} + 1$

$\mathrm{X}^{(\mathrm{t}+1)} \leftarrow \mathrm{X}^{(\mathrm{k}+1)}$

$\mathrm{D}_2^{(\mathrm{t}+1)} \leftarrow \mathrm{D}_2^{(\mathrm{k}+1)}$

7. Update iteration:

$\mathrm{t} \leftarrow \mathrm{t} + 1$

8. Stop: Until the pre-designed termination criteria are reached.

IV. Experiments and Analysis

In this part, the unmixing results obtained by our SCLRSU algorithm will be compared with those of SUNSAL [26], CLUNSAL [27], SUNSAL-TV [29], S^2SWSU [33], and SUSRLR-TV [50] through two simulated datasets and two widely used real HSIs. Four evaluation indexes, which are the signal reconstruction error (SRE) [59], the root mean square error (RMSE) [60], sparsity [61], and success probability (Ps) [62], are used to validate these algorithms. The calculation of SRE is shown as follows:

$$\mathrm{SRE} = 10\log_{10} \frac{E\left[\|X\|_{\mathrm{y}}^2\right]}{E\left[\|X - \hat{} X\|_{\mathrm{y}}^2\right]} \tag{18}$$

where X represents the true abundances of endmembers and $\hat{}X$ is its estimated value by using different SU methods. The larger the SRE value, the better the unmixing performances. RMSE is expressed as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{m}\sum_{j=1}^{m}\left(X_{ij} - X_{1ij}\right)^2} \tag{19}$$

where X_{1ij} and X_{ij} are the ground truth and the estimated abundances for a given endmember. The smaller the RMSE value, the better the unmixing performances.

Sparsity is expressed as follows:

$$\mathrm{Sparsity} = \frac{s}{mn} \tag{20}$$

where s is set as the number of abundance values greater than 0.005 in the estimation matrix X, and mn is the product of rows and columns in matrix X, that is, the total number of elements. The smaller the sparsity, the higher the sparsity of the estimated abundance matrix.

Ps is the probability that the relative error is less than a certain threshold. This measure is widely used in sparse regression.

Ps is expressed as follows:

$$\mathrm{Ps} = P\left(\frac{\|X\hat{}X\|^2}{\|X\|^2}\right) \le \mathrm{th} \tag{21}$$

where th is the threshold. For example, when th = 10, Ps = 1. This means that the relative error of all abundance vectors is less than 0.1. Ps represents the stability of abundance estimation. In this article, we think that the estimation result is accurate when $(\|X - \hat{} X\|^2/\|X\|^2)$ is smaller than 5.

A. Simulation Data Experiment

In this section, we use the spectral library named splib06 to conduct SU experiments. The library was released by the

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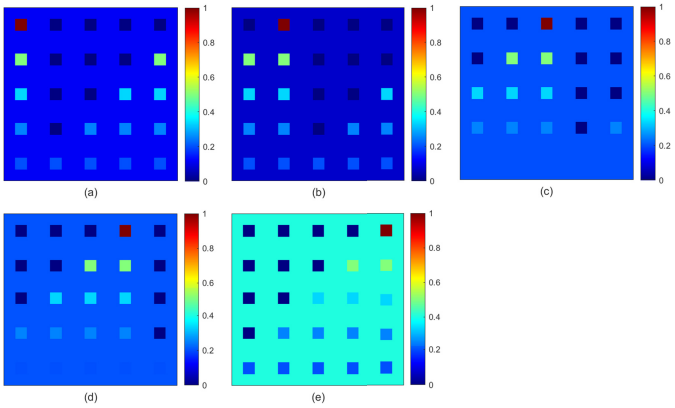


Fig. 2. True abundance of simulated image DS1. (a) Spectra of the five endmembers. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4. (e) Endmember 5.

United States Geological Survey (USGS) [63] in September 2007. The spectral library $A^{224*240}$ is composed of 240 different spectra, each with 224 spectral bands, and the spectral range is 0.4–2.5 nm.

Dataset 1 (DS1) is composed of five randomly selected spectral features from A following the LMM, including 75×75 pixels, and each pixel contains 224 bands. The abundance of DS1 is constrained by nonnegative and sum constraints. In DS1, some square areas are composed of pure pixels, while others are composed of two to five endmembers. The background pixels of the dataset consist of the same five endmembers, and their abundances are fixed to 0.1149, 0.0741, 0.2003, 0.2055, and 0.4051, respectively. Fig. 2 shows the true abundance of the five endmembers.

Dataset 2 (DS2) is generated by randomly selecting nine spectral signatures from A, which has 100×100 pixels. The abundances of the nine spectral signatures follow a probability Dirichlet distribution and are constrained by ANC and ASC constraints. As shown in Fig. 3, the abundances of DS2 show piecewise smoothness in their spatial distribution.

After generating DS1 and DS2, the Gaussian noise with different signal-to-noise ratios (SNRs) [64] (20, 30, and 40 dB) is added to the three datasets. According to the results of previous studies, the highest SRE values obtained by different algorithms are different. For SUNSAL and CLUNSAL algorithms, we select the most appropriate regular term λ from the sequence {0.0005, 0.001, 0.005, 0.01, 0.05, 0.5, 1, 5}. Similarly, according to previous experiments, we will select the most suitable parameters λ and λ_{TV} from the sequence: {0.0005, 0.001, 0.005, 0.01, 0.05, 0.5, 1, 5, 10}. Since both SUNSAL and CLUNSAL have only one regularization parameter, SUNSAL-TV, S2SWSU, and SUSRLR-TV have two regularization parameters, and it will take more time to select the most appropriate regularization parameter for SUNSAL-TV, S2SWSU, and SUSRLR-TV than for SUNSAL and CLUNSAL. In all algorithms, we fix the Lagrange parameter as $\mu = 0.1$. All the algorithms in this article are executed on a Windows 10 system and completed by MATLAB r2019b. The CPU size is 3.6 GHz, and the memory is 16 GB. Next, we will compare the unmixing effect of the

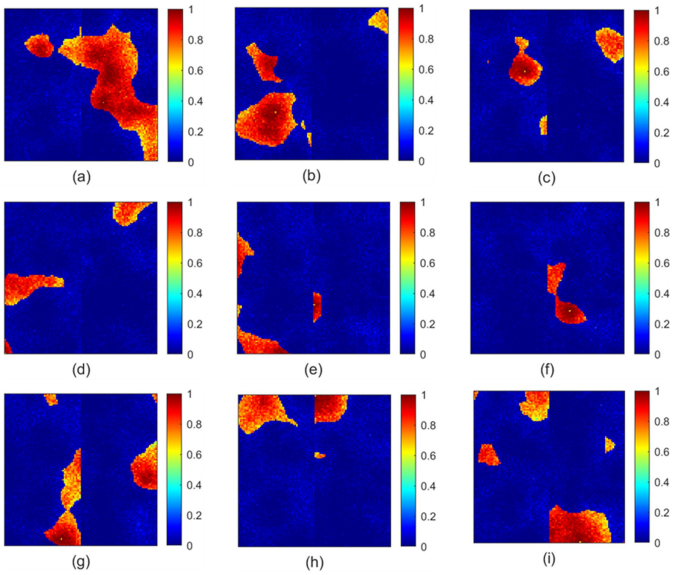


Fig. 3. True abundance of simulated image DS2. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4. (e) Endmember 5. (f) Endmember 6. (g) Endmember 7. (h) Endmember 8. (i) Endmember 9.

SCLRSU algorithm with that of SUNSAL [26], CLUNSAL [27], SUNSAL-TV [29], S2SWSU [33], and SUSRLR-TV [50] algorithms. Tables I and II show the SRE (dB), RMSE, sparsity, and Ps values obtained by some of the above algorithms when the SNRs are 20, 30, and 40 dB, respectively. At the same time, the optimal regularization parameters of the comparison algorithm are also provided in the following. In addition, in order to further intuitively compare the unmixing effect of various algorithms and accurately analyze the antinoise performance of the algorithm, we randomly select two endmembers in two datasets for experiments with SNR = 20, 30, 40, respectively. Figs. 4 and 5 show two representative estimated abundances of endmember 1 and endmember 3 in DS1, respectively. Figs. 6 and 7 show two representative estimated abundances of endmembers 5 and 9 in DS2, respectively.

From Tables I and II, it can be clearly seen that the SCLRSU algorithm proposed in this article obtains the highest SRE values than the other algorithms: SUNSAL [26], CLUNSAL [27], SUNSAL-TV [29], S2SWSU [33], and SUSRLR-TV [50] with DS1 and DS2. The above algorithms can obtain a larger SRE value at a higher SNR level (the difference is not obvious). Nevertheless, when the SNR level is reduced, the difference between the various algorithms is amplified. First, we notice that the quantitative results of other methods are much better than that of SUNSAL, which shows that the SU method taking spatial information into consideration can achieve satisfactory results in most cases. In addition, considering the different SNR values in the above table and the different characteristics of each algorithm, we can see that, for the SUNSAL and CLUNSAL algorithms (with only one regularization parameter), the SRE values obtained by the two algorithms (with SNR = 20 and 30) are lower than those of other algorithms. This is because the two algorithms have only one regularization parameter.

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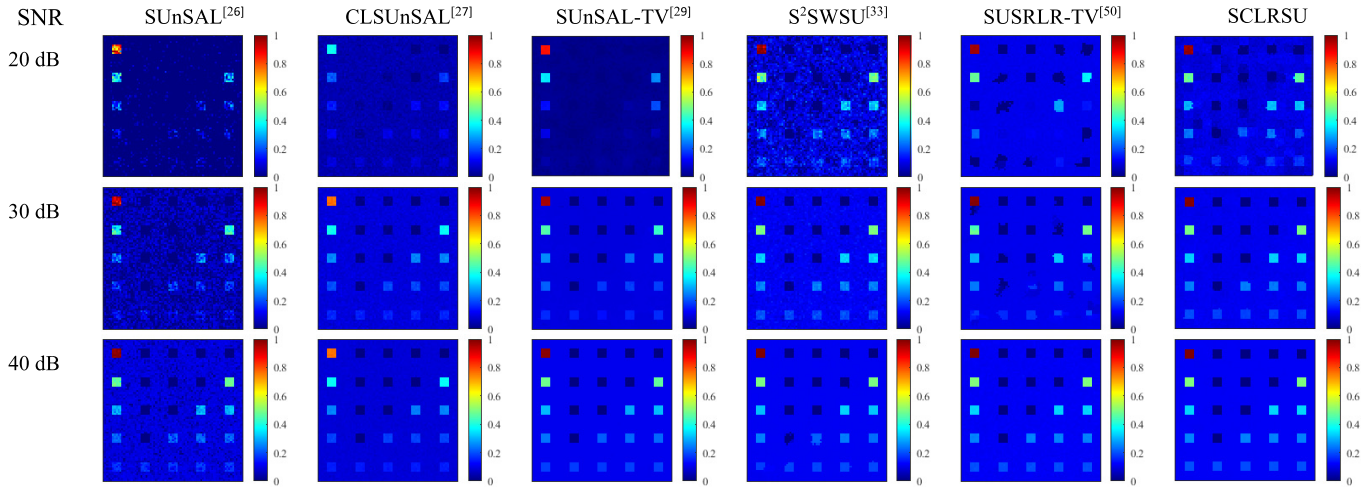


Fig. 4. Estimated abundance of endmember 1 in DS1 with different algorithms under various SNRs.

TABLE I
COMPARISON OF THE RESULTS OF VARIOUS ALGORITHMS (DIFFERENT SNRS) WITH DS1

SNR		SUnSAL [26]	CLSUnSAL [27]	SUnSAL-TV [29]	S^2SWSU [33]	SUSRLR-TV [50]	SCLRSU
20dB	SRE (dB)	4.26	5.40	10.97	9.36	13.37	**14.28**
	RMSE	**0.005**	0.02	0.015	0.018	0.010	0.010
	parameter	λ=0.1	λ=0.5	λ=0.005	λ=0.01	λ=0.01	λ=0.01
				λ_{TV}=0.05		λ_{CS}=0.05	λ_{CS}=0.05
	Sparsity	0.0523	0.0448	0.0845	0.06	0.0216	**0.0204**
	Ps	0.8158	0.8637	0.9612	0.9721	0.9735	**0.9774**
30 dB	SRE (dB)	10.45	13.35	17.80	21.71	22.73	**23.45**
	RMSE	0.016	0.023	0.007	0.004	0.003	**0.003**
	parameter	λ=0.01	λ=0.5	λ=0.005	λ=0.01	λ=0.01	λ=0.01
				λ_{TV}=0.005		λ_{CS}=0.05	λ_{CS}=0.05
	Sparsity	0.0622	0.0336	0.0354	0.0334	0.0207	**0.0201**
	Ps	0.9619	0.9743	0.9732	0.9825	0.9863	**0.9871**
40 dB	SRE (dB)	17.88	11.81	21.02	27.78	29.16	**30.52**
	RMSE	0.007	0.014	0.004	0.002	0.001	**0.001**
	parameter	λ=0.005	λ=0.005	λ=0.001	λ=0.001	λ=0.01	λ=0.01
				λ_{TV}=0.001		λ_{CS}=0.05	λ_{CS}=0.05
	Sparsity	0.0307	0.0226	0.0225	0.0208	0.021	**0.02**
	Ps	1	1	1	1	1	**1**

In addition, we notice that, when SNR = 20 dB, S^2SWSU obtains the lowest SRE value among the two regularized parameters, which indicates that the S^2SWSU algorithm is also sensitive to noise. However, when SNR = 30 and 40 dB, the

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TABLE II
COMPARISON OF THE RESULTS OF VARIOUS ALGORITHMS (DIFFERENT SNRS) WITH DS2

SNR		SUnSAL [26]	CLSUnSAL [27]	SUnSAL-TV [29]	S^2SWSU [33]	SUSRLR-TV [50]	SCLRSU
20dB	SRE (dB)	6.65	8.54	9.42	7.69	15.03	**15.10**
	RMSE	0.074	0.012	0.011	0.014	0.007	**0.006**
	parameter	λ=0.05	λ=1.00	λ=0.0005	λ=0.1	λ=0.01	λ=0.01
				λ_{TV}=0.05		λ_{CS}=0.05	λ_{CS}=0.05
	Sparsity	0.0456	0.0352	0.0689	0.0506	0.020	**0.018**
	Ps	0.5971	0.7438	0.7528	0.8647	0.8930	**0.9044**
30 dB	SRE (dB)	8.06	11.12	14.44	15.48	25.98	**26.06**
	RMSE	0.027	0.009	0.006	0.005	0.001	**0.001**
	parameter	λ=0.01	λ=1.00	λ=0.005	λ=0.005	λ=0.05	λ=0.05
				λ_{TV}=0.01		λ_{CS}=0.05	λ_{CS}=0.05
	Sparsity	0.0456	0.0352	0.0689	0.0506	0.019	**0.018**
	Ps	0.9314	0.9429	0.9732	0.9817	0.9854	**0.9967**
40 dB	SRE (dB)	10.88	13.66	22.09	28.23	29.75	**30.15**
	RMSE	0.013	0.007	0.002	0.001	0.001	**0.001**
	parameter	λ=0.005	λ=1.00	λ=0.005	λ=0.01	λ=0.01	λ=0.05
				λ_{TV}=0.005		λ_{CS}=0.05	λ_{CS}=0.05
	Sparsity	0.0456	0.0352	0.0689	0.0506	0.019	**0.018**
	Ps	0.9915	0.9938	0.9954	0.9981	0.9987	**1**

S^2SWSU algorithm gets better results than the SUNSAL-TV algorithm, and when SNR = 40, the S^2SWSU algorithm is obviously ahead of the SUNSAL-TV algorithm. It is not difficult to see that the SCLRSU algorithm achieves the highest accuracy on two datasets, no matter the noise level. It is particularly noteworthy that, when SNR = 30 and 40 dB, the SRE value of the SCLRSU algorithm is about 2–4 dB higher than that of the SUNSAL-TV algorithm and the S^2SWSU algorithm (when SNR = 30 and 40 dB). In addition, we can also notice that, when the noise is high (such as SNR = 20 dB), the SRE values obtained by the SCLRSU algorithm on the two datasets are about 5 dB higher than those obtained by other algorithms. Because the local regions share the same active endmember set and keep the local cooperative sparsity of the abundance matrix, SCLRSU obtains higher SRE than CLUNSAL. Finally, it can be seen from the table that the SCLRSU algorithm achieves the best results in all noise conditions and has high sparsity and stability. This is because SCLRSU can adaptively segment the image into uniform regions to maintain local synergy.

From Figs. 4 and 5, we can see that the abundance maps obtained by all algorithms are poor when the noise is high, and the results gradually improve with the gradual reduction of noise. It is obvious that, since SUNSAL does not consider the spatial information, the worst solution is obtained by this algorithm on both datasets. For the SUNSAL-TV algorithm, we can clearly see that, due to the consideration of spatial consistency, the abundance maps obtained by SUNSAL-TV under various noise levels exhibit clear square boundaries, so this algorithm is similar to other algorithms. For DS2, we can draw similar conclusions from Figs. 6 and 7. In this article, superpixel segmentation is used to mine reliable spatial information, and the local correlation of the abundance matrix is fully mined by cooperating with low-rank constraints to obtain more accurate abundance maps. To sum up, we can clearly see that fully mining local spatial information can obtain higher quality abundance maps and significantly improve the unmixing accuracy. In general, our algorithm obtains the abundance map closest to the true value of the ground from the above two

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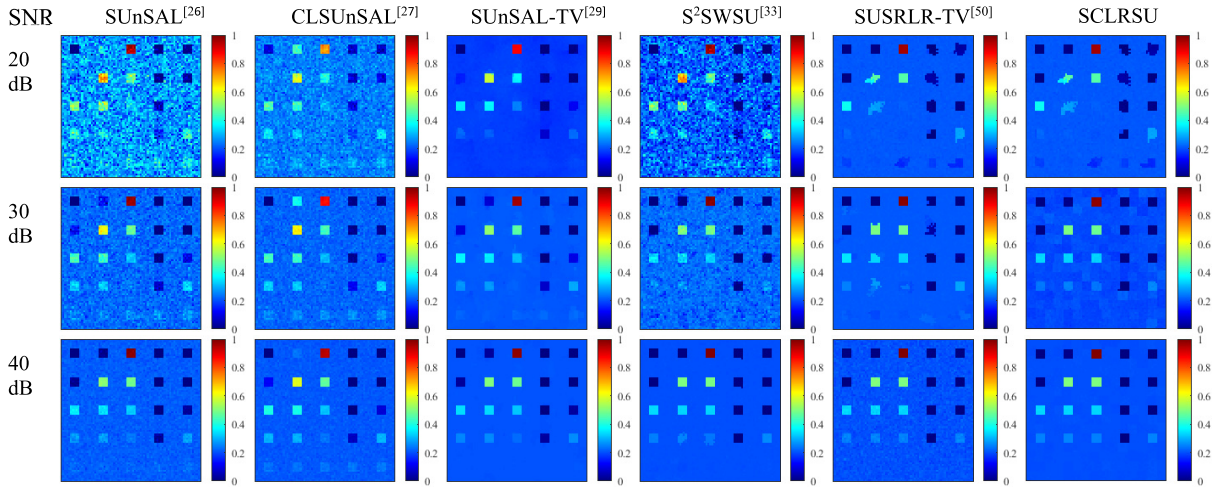


Fig. 5. Estimated abundance of endmember 3 in simulated DS1 with different algorithms under various noise levels.

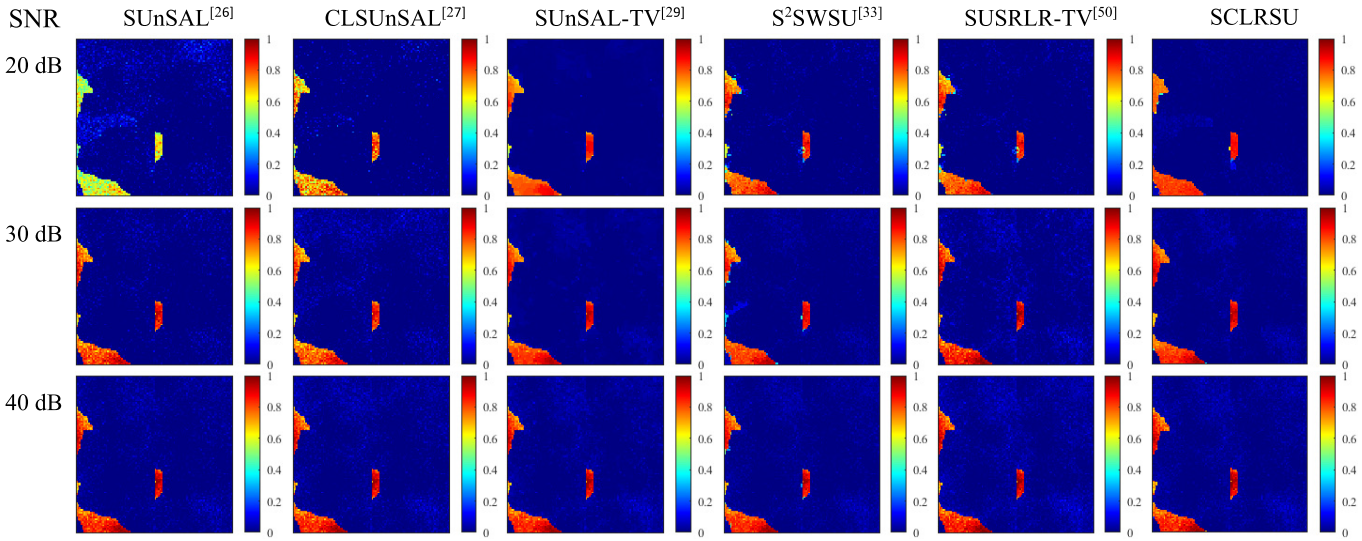


Fig. 6. Estimated abundance of endmember 5 in DS2 with different algorithms under various noise levels.

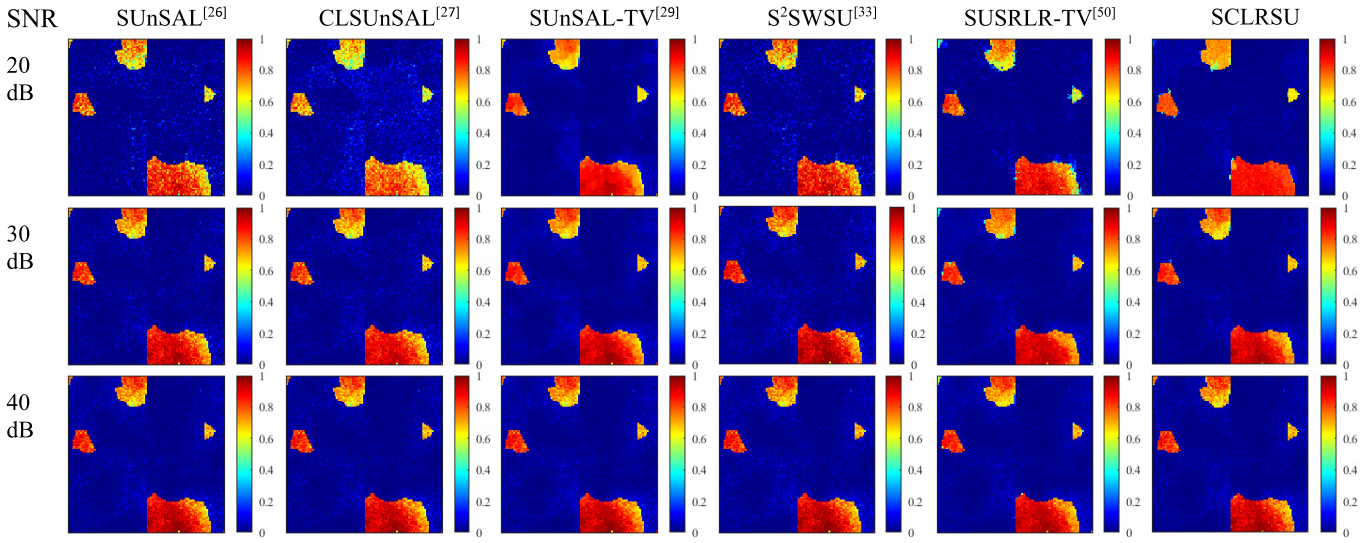


Fig. 7. Estimated abundance of endmember 9 in simulation DS2 with different algorithms under various noises.

groups of simulated data, and the conclusion drawn from the obtained maps is consistent with the data recorded in Tables I and II.

B. Real Data Experiments

For the purpose of further verifying the performance of the proposed SCLRSU, we test it on the actual hyperspectral

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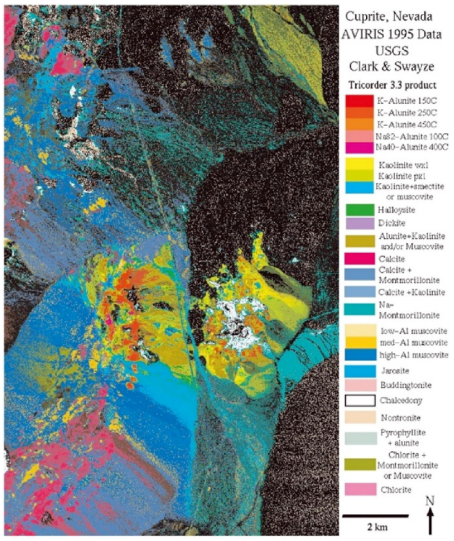


Fig. 8. USGS map showing the presence of minerals in Cuprite.



Fig. 9. HYDICE Urban hyperspectral data.

data. We use the famous airplane visible imaging spectrometer (AVIRIS) Cuprite [65] (see Fig. 8) and Urban datasets [66] (see Fig. 9).

1) The Cuprite dataset was collected by AVIRIS sensor in June 1997 and is widely used in hyperspectral classification and unmixing. There are 224 bands in the Cuprite dataset with a wavelength range of 0.4–2.5 μm and a spectral resolution of 10 nm, covering the Cuprite Mining area in Nevada. We use the subimage of 250 $\times$ 191 pixels in our experiment. In this area, there are mainly minerals such as alunite, kaolinite, chalcedony, and buddingtonite, all of which have pure pixels. In addition, it also includes a small number of anomalous minerals, such as buddingtonite, nontronite, halloysite, and dickite. Due to the fact that some spectral bands have water absorption and low SNR values, 188 spectral bands were used after removing some spectral bands. In this experiment, the spectral dictionary library $A \in R^{188\times240}$ is generated from the spectral library (splib06a) of USGS. In Fig. 10, three minerals, alunite, buddingtonite, and chalcedony, are selected as instances.
2) The Urban dataset was taken by HYDICE hyperspectral camera in October 1995 in the suburb of Texas, USA. The original HSI contains 94 249 (307 $\times$ 307) pixels and consists of 210 spectral channels, covering the spectral range of the electromagnetic spectrum from 0.4 to 2.5 μm with a spectral resolution of 10 nm. After deleting the noise channels, images mainly contain four kinds of land: asphalt road, roof, grass, and trees.

Fig. 10 shows the estimated abundance maps of three different endmembers (alunite, ammonite, and chalcedony) by SUNSAL, CLUNSAL, SUNSAL-TV, S2SWSU, SUSRLR-TV, and the proposed SCLRSU. The parameter setting of each algorithm is the same as that of the simulation data experiment. We can see from Fig. 10 that the positions of these three minerals can be identified by all methods, which proves that the SU method is effective for real HSIs. However, we can still find a few subtle differences in the obtained abundance map. The main disadvantage of SUnSAL is that, compared with other algorithms, it obtains lower abundance estimates. Especially for chalcedony, distinguishing this mineral from the background is difficult for SUnSAL. For the fractional abundance estimated by SUnSAL-TV, the excessive smoothing of alunite and ammonite feldspar is obvious. In addition, the algorithm proposed in this article and the S^2SWSU algorithm become closer to the reference results, specifically for alunite. The method has good performance for alunite, and to a certain extent, it avoids the phenomenon of excessive smoothing. At the same time, we also find that, compared with SUnSAL, CLUNSAL, SUNSAL-TV, and S^2SWSU, the estimated abundance values given by the algorithm in the areas divided into different minerals are usually higher or comparable. The results show that the proposed algorithm is effective for unmixing the actual hyperspectral data.

In order to further analyze the performance of various algorithms for real data unmixing, the effectiveness of the proposed algorithm was verified by using the Urban dataset. In this dataset, we selected four end-member experiments, and Fig. 11 shows the estimated abundance maps of four different endmembers (roads, grasslands, trees, and roofs) by SUnSAL, CLUNSAL, SUNSAL-TV, S2SWSU, SUSRLR-TV, and our algorithm. The parameter setting of each algorithm is the same as that of the simulation data experiment. We can see from Fig. 11 that each algorithm can effectively extract the four endmembers, and the above results further show the effectiveness of the sparse decomposition method for real data. From the estimated abundance map, we can clearly see that, compared with other algorithms, the SUnSAL algorithm is better in extracting grassland, but the extraction results of other endmembers are not ideal. Especially for roads, the abundance map results estimated by the SUnSAL algorithm are difficult to distinguish from the background. In addition, we noticed that the result of grassland extraction by SUnSAL-TV was oversmooth to some extent. In addition, from the synthesis of the four endmembers, the results of the proposed algorithm and the S^2SWSU algorithm are closer to the reference. Compared with SUnSAL, CLUNSAL, SUNSAL-TV, S2SWSU, and SUSRLR-TV, the estimated

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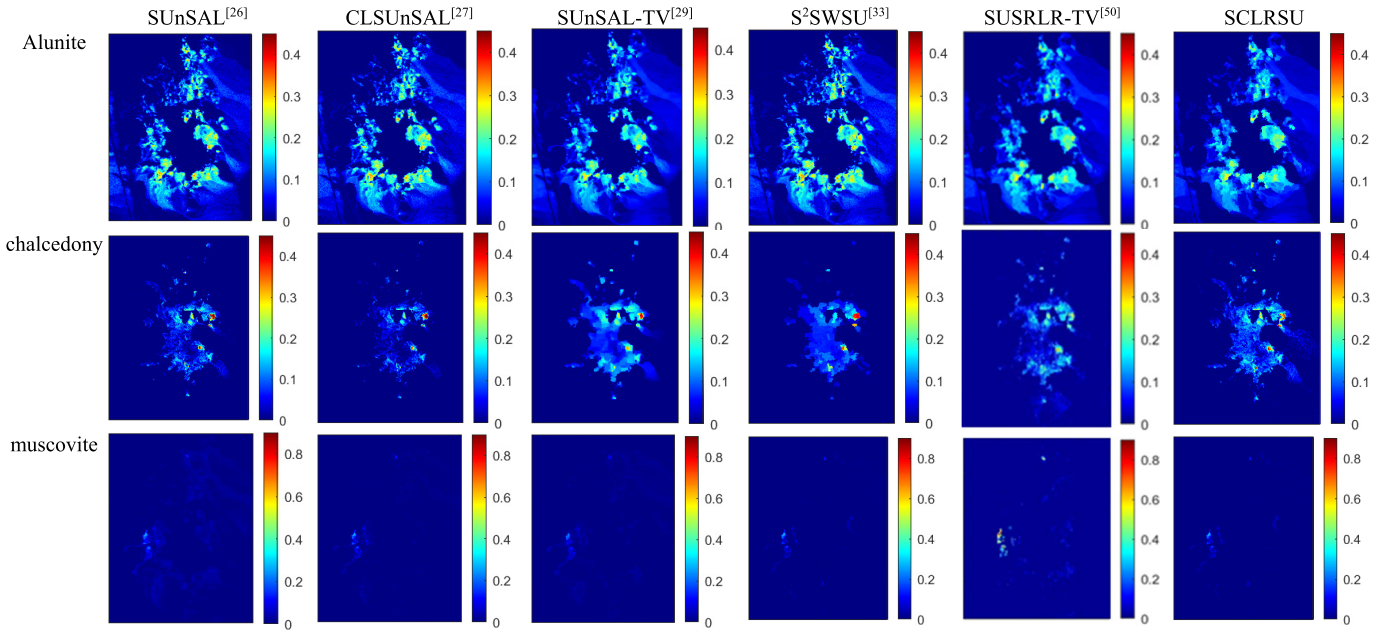


Fig. 10. Estimated abundance map of three different endmembers for the Cuprite dataset.

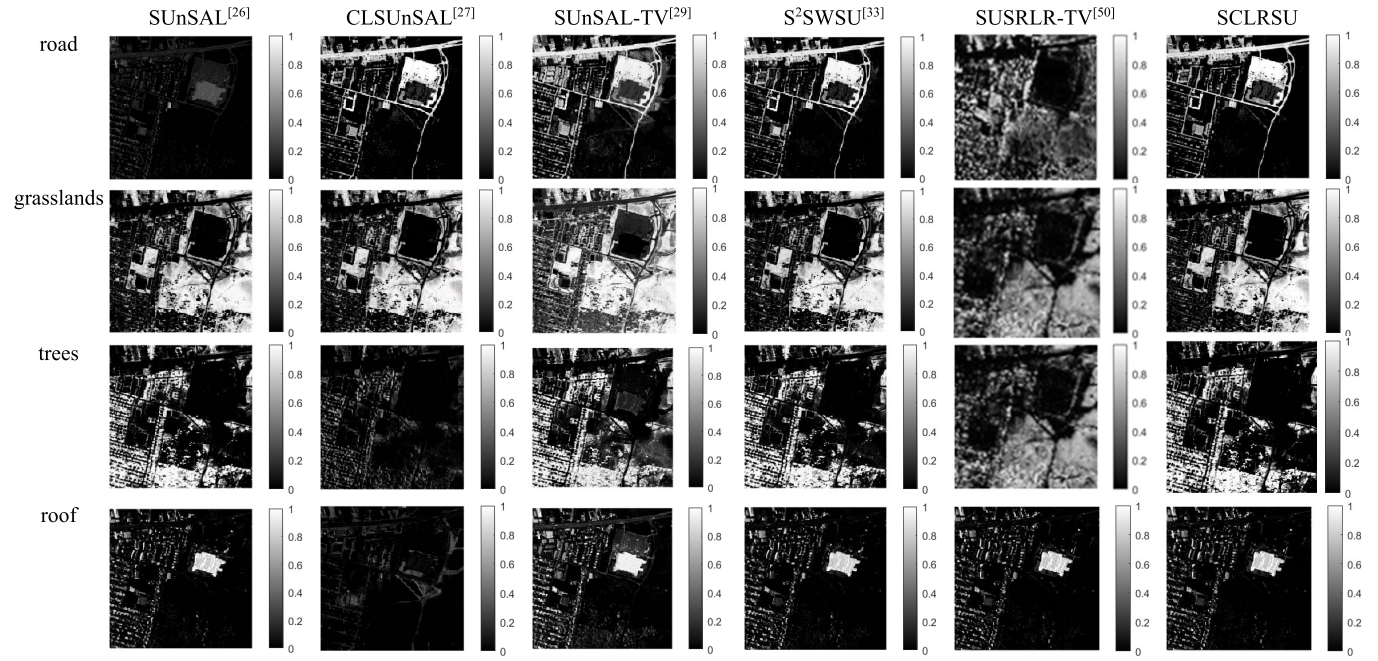


Fig. 11. Estimated abundance map of four different endmembers in the Urban dataset.

abundance of the four endmembers is usually higher or comparable. Combining the results from the above two datasets, we can draw the conclusion that the algorithm in this article can get better results in unmixing the actual hyperspectral data.

C. Parameter Change Analysis

In the following, we focus on analyzing the influence of the change of parameters λ and λcs on SRE in the SCLRSU algorithm proposed in this article. From the expressions mentioned above, it is not difficult to find that the parameters of λ and λcs control the two kinds of regularization constraints, so the selection of parameters has an important influence on the performance of this algorithm. Specifically, λ is used to extract spatial information and spectral information of the abundance map, while λcs plays an important role in the coordination and correlation of pixels in the local area. In order to accurately analyze the influence of the two parameters on the unmixing results, all combinations of λ and λcs are considered, the corresponding parameter combination when SRE (dB) reaches the maximum value is selected, and the influence of three different SNRs on the two parameters is

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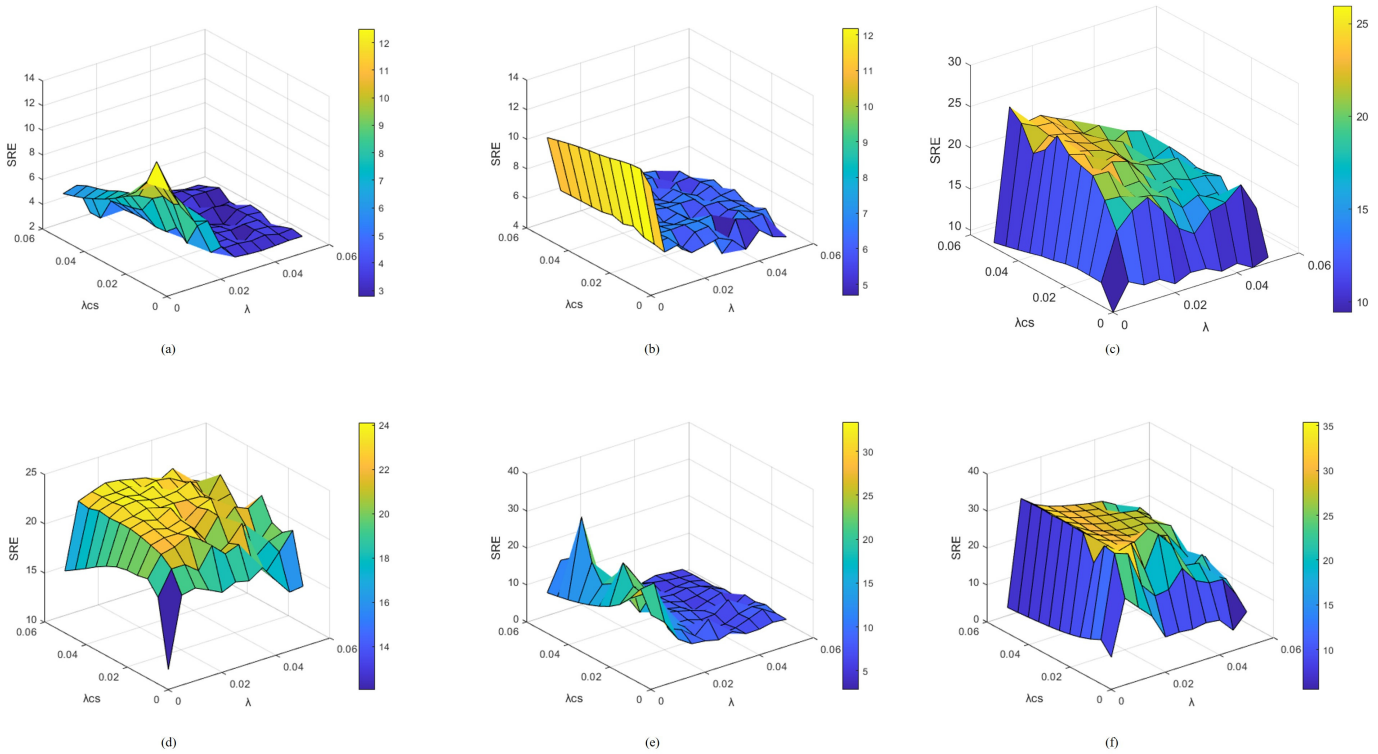


Fig. 12. Influence of the change of λ and λcs on SRE (dB) under different noise levels. (a) DS1 with SNR = 20. (b) DS1 with SNR = 30. (c) DS1 with SNR = 40. (d) DS2 with SNR = 20. (e) DS2 with SNR = 30. (f) DS2 with SNR = 40.

TABLE III
TIME FOR SINGLE ITERATION OF DIFFERENT ALGORITHMS AT DIFFERENT SNRS (dB)

SNR	SUnSAL	CLSUnSAL	SUnSAL-TV	S2SWU	SCLRSU
20 dB	0.001	0.002	0.283	0.262	0.359
30 dB	0.001	0.002	0.299	0.261	0.357
40 dB	0.001	0.002	0.296	0.260	0.352

also considered. We will set λ and λcs as [0, 5e-3, 1e-3, 5e-2, 1e-2, 5e-1, 1e-1].

Fig. 12 shows the influence of the change of λ and λcs on SRE (dB) values when there are different noise levels in DS1 and DS2. From Fig. 12, we can see that the parameter λcs has a great influence on the model, while λ has little influence on the model.

In addition, we analyze the running time of the five methods considered in this work and report the running time required by various methods by averaging several experiments at different SNRs on DS1. Table III shows the running time of the five unmixing methods. Experimental results show that the SCLRSU method proposed in this article takes the most time because it needs to segment the whole HSI to obtain local blocks and solve the abundance map, which is the most complex of the above methods. As for the change of SNR, we find that it has little effect on the running time of the algorithm. In addition, we noticed that, although the time of the proposed method is obviously longer than the traditional algorithm, and the running time is similar to that of S^2SWSU, the most accurate unmixing result can be obtained by the method introduced in this article, and the reconstructed abundance images are the closest to the reference abundance maps among the five algorithms.

D. Superpixel Change Analysis

In addition to the two parameters of the model proposed in this chapter, we also analyze the influence of segmentation on the unmixing results. We try to find the optimal number of superpixels by trial and error. Similar to the above experimental methods, we also use three SNR experiments to analyze the influence of the number of superpixel segmentation on the unmixing results under different SNR conditions. Fig. 13 shows the influence of the number of superpixels on the SRE (dB) in DS1 and DS2. From Fig. 13, it is not difficult to see that, if the number of superpixels is too small or too large, it is difficult to get satisfactory results. When SNR = 20 or 30, the SRE (dB) results drop sharply when the number of DS1 superpixels is more than 260, and when SNR = 40, the number of DS1 superpixels changes at 270. This reveals that the SNR has a certain influence on the optimal number of superpixels. The critical values of DS2 are 460 and 470, respectively. Therefore, the number of superpixels used for DS1 and DS2 is 260 and 460, respectively.

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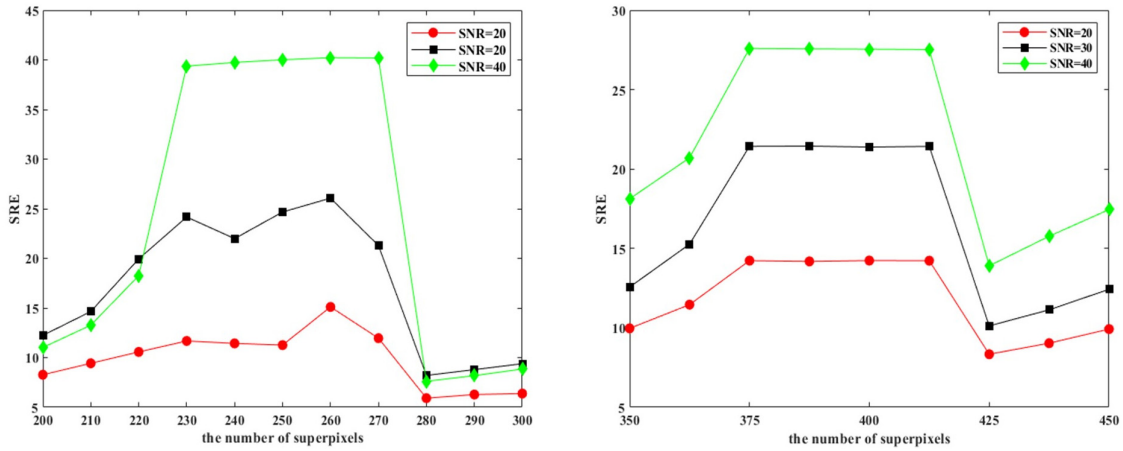


Fig. 13. Influence of superpixel number on SRE values under different SNRs.

V. Conclusion

In order to better characterize local spatial information in SU, this article proposes a new SCLRSU algorithm that exploits superpixels. Here, SLIC is used to obtain the local similar regions of the HSI, and the spatial–spectral information is considered at the same time. In addition, a collaborative sparsity constraint is used in similar superpixels to extract spectral correlation, and a low-rank constraint is also added. The ADMM is used to finally solve the problem. The effectiveness and superiority of the proposed method are verified by two sets of simulated data and two sets of real data. Compared with traditional algorithms, our experiments show that the proposed SCLRSU algorithm performs both qualitatively and quantitatively better. More efficient implementations will be carried out in our future studies.

Appendix

In this appendix, we will present the process of how to solve the SCLRSU model. To solve the problem in (17), the ADMM strategy is adopted. The ADMM algorithm aims to transform a complex problem that is difficult to solve into several easy-to-solve subproblems by introducing multiple variables and solve these subproblems separately to solve this problem. By introducing variables $V_1, V_2, V_{3,r}$, and V_4, the model can be rewritten as

$$\begin{aligned}\min\ & \frac{1}{2}||V_1 - Y||_F^2 + \lambda\left\|\left(W_{\text{spe}}W_{\text{spa}}\right)\odot V_2\right\|_{2,1} \\ & + \lambda \text{cs}\sum_{r=1}^{R}||V_{3,r}||_{21,*} + l_{R+}(V_4) \\ \text{s.t. } & V_1 = AX \\ & V_2 = X \\ & V_{3,r} = X_r \\ & V_4 = X. \end{aligned} \tag{22}$$

For X, the solution is

$$\begin{aligned}X^{k+1} = \left(A^TA + 2I\right)^{-1}\Big(A^T\Big(V_1^{(k)} + D_1^{(k)}\Big) \\ + V_2^{(k)} + D_2^{(k)} + V_{3,r}^{(k)} + D_{3,r}^{(k)}\Big)\end{aligned} \tag{23}$$

where A^T be a transposition of A and I is its identity matrix.

Next, we need to solve each variable in the step-in turns. Calculate V_1. The solution V_1 problem can be written as

$$\min\ \frac{1}{2}||V_1 - Y||_F^2 + \frac{\mu}{2}||V_1 - AX + D_1||_F^2. \tag{24}$$

The solution is

$$\begin{aligned}V_1^{(k+1)} = \frac{1}{1+\mu}\Big(Y + \mu\Big(AX^{(k+1)} - D_1^{(k)}\Big)\Big)\min\frac{1}{2}||V_1 - Y||_F^2 \\ + \frac{\mu}{2}||V_1 - AX + D_1||_F^2.\end{aligned} \tag{25}$$

For V_2, the optimization problem can be written as

$$\begin{aligned}\lambda||W_{\text{spe}}W_{\text{spa}}\odot V_2|| + \frac{\mu}{2}||V_2 - X + D_2|_F^2\min\frac{1}{2}||V_1 - Y||_F^2 \\ + \frac{\mu}{2}||V_1 - AX + D_1|_F^2.\end{aligned} \tag{26}$$

For V_2, the solution to the optimization problem can be obtained by the vect-soft threshold method

$$V_2^{(k+1)} = \text{vect} - \text{soft}\left(X^{(k+1)} - D_2^{(k)}, \frac{\lambda}{\mu}W_{\text{spe}}^{(t)}W_{\text{spa}}^{(t)}\right) \tag{27}$$

in which (vect − soft threshold) is the line soft threshold function.

In order to obtain $V_{3,r}$, the following optimization problems need to be solved:

$$\lambda \text{cs}\sum_{r=1}^{R}||V_{3,r}||_{1/2,*} + \frac{\mu}{2}||V_{3,r} - X + D_{3,r}|_F^2. \tag{28}$$

For $V_{3,r}$, the solution can be obtained by the soft threshold method

$$V_{3,r}^{(k+1)} = \text{soft}\left(X^{(k+1)} - D_{3,r}^{(k)}, \frac{\lambda\text{cs}}{\mu}\right). \tag{29}$$

For V_4, the optimization problem is

$$l_{R+}(V_4) + \frac{\mu}{2}||V_4 - X + D_4|_F^2. \tag{30}$$

The solution of V_4 is

$$V_4^{(k+1)} = \max\left(X^{(k+1)} - D_4^{(k)}, 0\right). \tag{31}$$

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Through the above algorithm, we can extract the regions with similar local distribution structures in the endmembers and then unmix them based on these regional blocks. Because the endmember information in these similar regions is much smaller than the endmember information in the whole HSI and the endmember information in the library, according to sparse theory, the corresponding abundance will be sparser.

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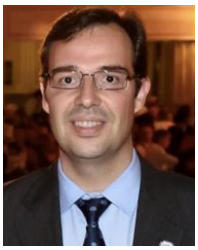
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